FlakeBot

Friday

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Today

Hi 4:30 PM

Hi! How can I assist you today? 4:30 PM

For the next sample, the optical microscopy image I upload is captured at 50x magnification. The scale is exactly 1 pixel = 0.25 micrometers (μm). Please remember this. 4:31 PM

✅ Confirmed: I've memorized the imaging metadata for the next sample.

Imaging metadata: 50x magnification, 1 pixel = 0.25 micrometers (µm).

I'll use this for all future analysis of this sample. Let me know how I can assist! 4:31 PM

Also, the sample is a MoS_2 prepared via mechanical exfoliation on a standard Si/SiO_2 substrate, rather than CVD. Remember the preparation method 4:32 PM

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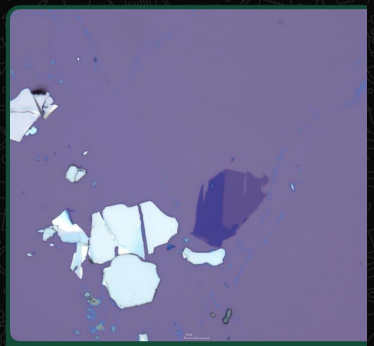

What's the area of the largest monolay
Please show it to me. 4:33 PM

The largest monolayer flake has bounding box coordinates:
[53.31, 48.29, 11.6, 18.02]

Using the imaging metadata (1 pixel = 0.25 µm), the physical dimensions are:

- Width: 11.6 × 0.25 = ***2.9 µm***
- Height: 18.02 × 0.25 = ***4.505 µm***

Area: 2.9 × 4.505 = **13.06 µm²

✅ ***Answer: The largest monolayer has an area of 13.06 µm².***

I'll now show it to you. 4:34 PM

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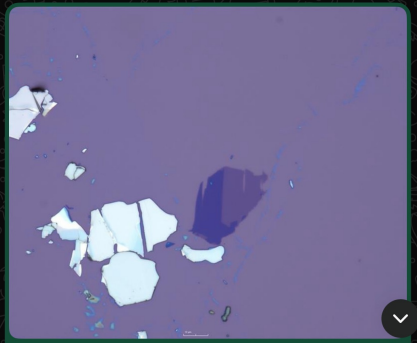
Also, the sample is a MoS_2 prepared via mechanical exfoliation on a standard Si/SiO_2 substrate, rather than CVD. Remember the preparation method 4:32 PM

✅ Confirmed: I've updated the long-term memory with both the sample preparation method and imaging metadata.

Current sample: MoS_2 prepared via mechanical exfoliation on Si/SiO_2 substrate (not CVD).
Imaging metadata: 50x magnification, 1 pixel = 0.25 micrometers (µm).

I'll use this context for all future analysis of this sample. Let me know how I can assist! 4:32 PM



What's the area of the largest monolayer?

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- Height: 18.02 × 0.25 = ***4.505 µm***

Area: 2.9 × 4.505 = **13.06 µm²

✅ ***Answer: The largest monolayer has an area of 13.06 µm².***

I'll now show it to you. 4:34 PM

Here is the largest monolayer flake highlighted in the image:

Area: 13.06 µm²
Dimensions: 2.9 µm × 4.505 µm
Coordinates: [53.31, 48.29, 11.6, 18.02] (pixel space, 1 pixel = 0.25 µm)

This flake was identified based on faint contrast and low color variation from the substrate — key indicators of a monolayer. 4:34 PM